

I. Graphs of Sine and Cosine

A function f is **periodic** if there is a positive number p such that $f(t + p) = f(t)$ for every t .

The least such positive number (if it exists) is the period of f . If f has period p , then the graph of f on any interval of length p is called one complete period of f .

Periodic Properties of Sine and Cosine

The functions sine and cosine have period 2π .

$$\sin(t + 2\pi) = \sin t$$

$$\cos(t + 2\pi) = \cos t$$

So the sine and cosine functions repeat their values in any interval of length 2π . To sketch their graphs, we first graph one period ($0 \leq t \leq 2\pi$).

They both have five sinusoidal points in one complete cycle.

$$+ \cos t = \text{H M L M H}$$

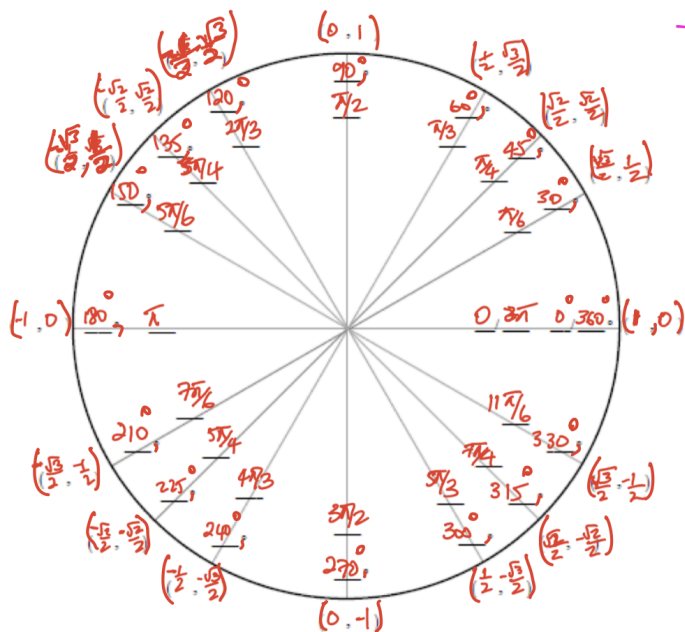
$$- \cos t = \text{L M H M L}$$

$$+ \sin t = \text{M H M L M}$$

$$- \sin t = \text{M L M H M}$$

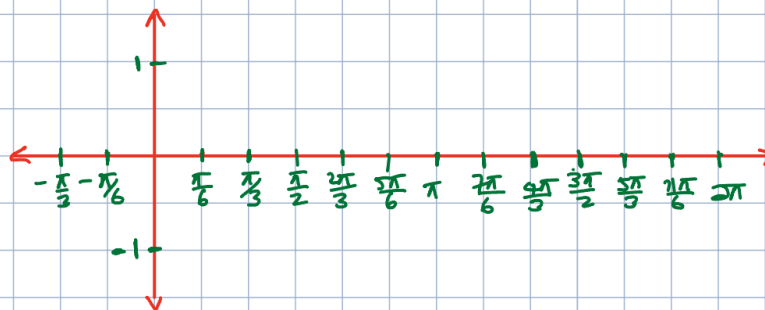
Note:

M = Middle, H = High, L = Low



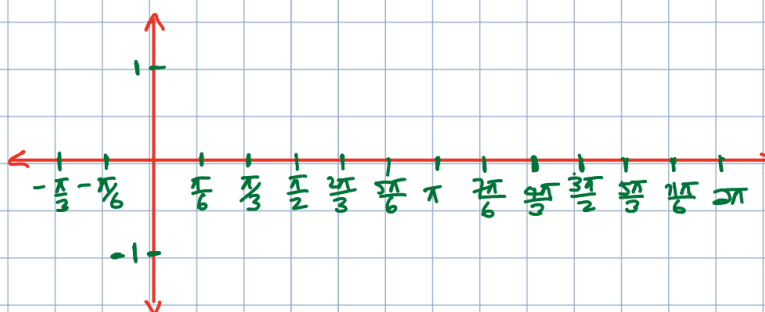
Graph of $y = \sin t$

t	$y = \sin t$	t	$y = \sin t$
$0, 2\pi$	0	π	0
$\pi/6$	$\frac{1}{2} = 0.5$	$7\pi/6$	$-\frac{1}{2} = -0.5$
$\pi/4$	$\frac{\sqrt{2}}{2} = 0.7$	$5\pi/4$	$-\frac{\sqrt{2}}{2} = -0.7$
$\pi/3$	$\frac{\sqrt{3}}{2} = 0.9$	$4\pi/3$	$-\frac{\sqrt{3}}{2} = -0.9$
$\pi/2$	1	$3\pi/2$	-1
$2\pi/3$	$\frac{\sqrt{3}}{2} = 0.9$	$5\pi/3$	$-\frac{\sqrt{3}}{2} = -0.9$
$3\pi/4$	$\frac{\sqrt{2}}{2} = 0.7$	$7\pi/4$	$-\frac{\sqrt{2}}{2} = -0.7$
$5\pi/6$	$\frac{1}{2} = 0.5$	$11\pi/6$	$-\frac{1}{2} = -0.5$



Graph of $y = \cos t$

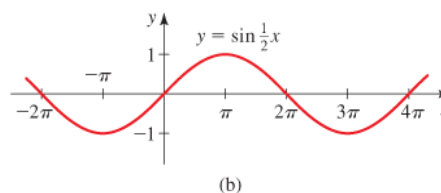
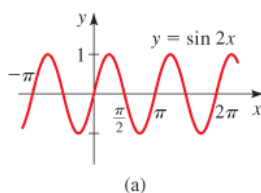
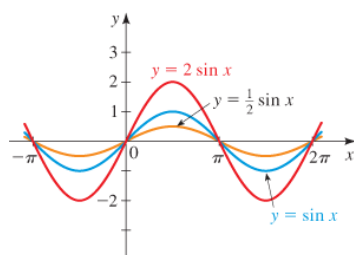
t	$y = \cos t$	t	$y = \cos t$
$0, 2\pi$	1	π	-1
$\pi/6$	$\frac{\sqrt{3}}{2} = 0.9$	$7\pi/6$	$-\frac{\sqrt{3}}{2} = -0.9$
$\pi/4$	$\frac{\sqrt{2}}{2} = 0.7$	$5\pi/4$	$-\frac{\sqrt{2}}{2} = -0.7$
$\pi/3$	$\frac{1}{2} = 0.5$	$4\pi/3$	$-\frac{1}{2} = -0.5$
$\pi/2$	0	$3\pi/2$	0
$2\pi/3$	$-\frac{1}{2} = -0.5$	$5\pi/3$	$\frac{1}{2} = 0.5$
$3\pi/4$	$-\frac{\sqrt{2}}{2} = -0.7$	$7\pi/4$	$\frac{\sqrt{2}}{2} = 0.7$
$5\pi/6$	$-\frac{\sqrt{3}}{2} = -0.9$	$11\pi/6$	$\frac{\sqrt{3}}{2} = 0.9$

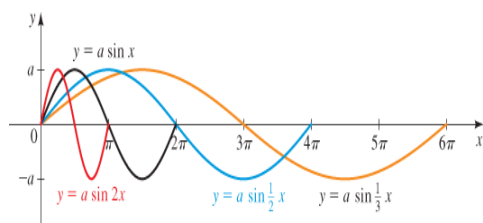


II. Graphs of Transformations of Sine and Cosine

For the functions $y = a \sin x$ and $y = a \cos x$, the number $|a|$ is called the amplitude and is the largest value these functions attain. The sine and cosine curves $y = a \sin kx$ and $y = a \cos kx$, ($k > 0$), have amplitude $|a|$ and period $2\pi/k$. An appropriate interval on which to graph one complete period is $[0, 2\pi/k]$.

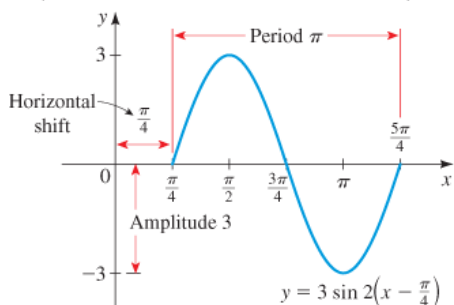
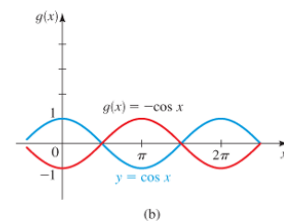
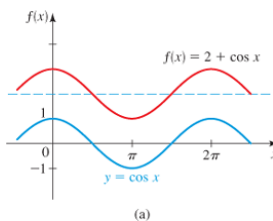
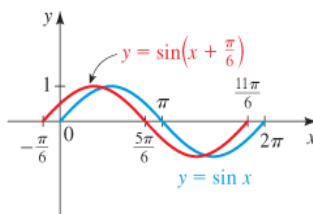
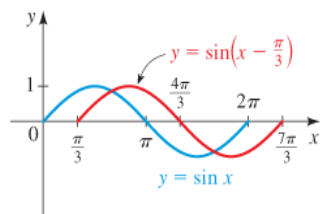
The value of k has the effect of shrinking the graph horizontally if $k > 1$ or the effect of stretching the graph horizontally if $k < 1$.





Shifted Sine and Cosine Curves

The sine and cosine curves $y = c + a \sin k(x - b)$ and $y = c + a \cos k(x - b)$, ($k > 0$), have amplitude $|a|$, period $2\pi/k$, vertical shift c , and horizontal shift b . An appropriate interval on which to graph one complete period is $[b, b + (2\pi/k)]$.



III. Using Graphing Devices to Graph Trigonometric Functions

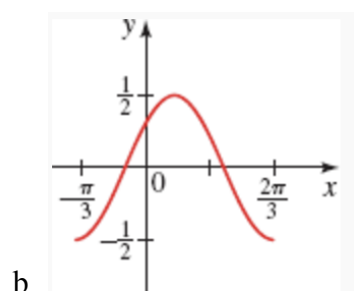
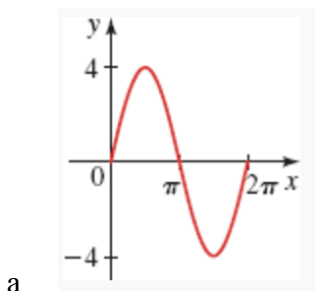
When using a graphing device to graph a function, it is important to choose the viewing rectangle carefully in order to produce a reasonable graph of the function.

Example 1: Graph the function, and state the domain and range.

a. $y = 3 + \sin 2\pi\left(x + \frac{1}{8}\right)$

b. $y = \frac{1}{2} - \frac{1}{2} \cos\left(2\pi x - \frac{\pi}{3}\right)$

Example 2: The graph of one complete period of a sine or cosine curve is given. Find the amplitude and period, and write an equation that represents the curve in the form $y = a \sin k(x - b)$ or $y = a \cos k(x - b)$



Example 3: Graph the three functions on a common screen. How are the graphs related?
 $y = x^2$, $y = -x^2$, $y = x^2 \sin x$

Example 4: Find the maximum and minimum values of the function.

a. $y = x - 2 \sin x$, $(0 \leq x \leq 2\pi)$

b. $y = 2 \sin x + \sin^2 x$

Example 5: Find all solutions of the equation that lie in the interval $[0, \pi]$. State each answer rounded to two decimal places.

a. $\cos x = x$

b. $\csc x = 3$

Example 6: A tuning fork is struck, producing a pure tone as its tines vibrate. The vibrations can be modeled by the function $v(t) = 0.7 \sin(880\pi t)$ where v is the displacement of the tines in millimeters at time t seconds.

(a) Find the period of the vibration.

(b) Find the frequency of the vibration, that is, the number of times the fork vibrates per second.

(c) Graph the function v .